MIL-T-45348(Ord)
11 January 1960
SUPERSEDING
MIL-T-11603A(Ord)(in part
4 December 1951

MILITARY SPECIFICATION

TRANSMISSION, CROSS-DRIVE; CD-500 SERIES

1. SCOPE

1.1 Scope. - This specification covers CD-500 series cross-drive transmissions designed for use in tracked, combat vehicles.

1.2 Classification. - Transmissions furnished under this specification shall be of the following types:

Type I	-	CD-500-3
Type II	-	CD-500-4

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS
Military

MIL-L-2104	-	Lubricating Oil, Internal-Combustion Engine, Heavy Duty.
MIL-L-10295	-	Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-T-10693	-	Transmissions, Cross-Drive, CD-500-3, CD-500-4, and CD-850-5; Preparation for Shipment and Storage of.
MIL-L-21260	-	Lubricating Oil, Internal-Combustion Engine, Preservative.

STANDARDS
Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of U.S. Military Property.

DRAWINGS
Ordnance Corps

K7527947	-	Transmission Assembly for CD-500-3.
K7707400	-	Transmission Assembly for CD-500-4.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction model. - The transmissions furnished under this specification shall be products which have been represented by a preproduction model that has been tested and has passed the preproduction model tests specified herein (see 4.3 and 6.1).

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3.2 Design and construction. - The design and construction shall be to the form and dimensions specified on Ordnance drawing K7527947 or K7707400, as applicable.

3.2.1 Transmission oil. - Transmission oil conforming to grade 1 of Specification MIL-L-21260 shall be used during transmission testing and production runs, except that oil conforming to Specification MIL-L-10295 shall be used when transmission is operated in ambient temperature below 0°F.

3.2.1.1 Service. - Transmission oil conforming to grade 10 of Specification MIL-L-2104 shall be used after installation of the transmission in its intended vehicle.

3.3 Performance.

3.3.1 Pressure requirements. - When tested as specified in 4.6.2, the transmission shall meet the requirements specified in 3.3.1.1 through 3.3.1.5.

3.3.1.1 Operating torque and pressure. - When tested as specified in 4.6.2.6, the input torque, main oil pressure, lubricating oil pressure, converter "in" oil pressure and the converter "out" oil pressure shall conform to the requirements specified in table I.

Table I - Operating torque and pressure

Item	Type I		Type II	
	Min.	Max.	Min.	Max.
Input torque - foot-pounds	65	92	57	91
Main oil pressure - p.s.i.	136	169	143	163
Lubricating oil pressure - p.s.i.	39	49	39	49
Converter "in" oil pressure - p.s.i.	70	96	70	96
Converter "out" oil pressure - p.s.i.	70	101	70	101

3.3.1.2 Right hand and left hand steer pressure. - When tested as specified in 4.6.2.2, the right hand and left hand steer pressure shall be not more than 5 p.s.i. below the main oil pressure.

3.3.1.3 High gear clutch oil pressure. - When tested as specified in 4.6.2.3, the high gear clutch oil pressure shall be not more than 5 p.s.i. below the main oil pressure.

3.3.1.4 Low gear clutch oil pressure. - When tested as specified in 4.6.2.4, the low gear clutch oil pressure shall be not more than 5 p.s.i. below the main oil pressure.

3.3.1.5 Reverse gear clutch oil pressure. - When tested as specified in 4.6.2.5, the reverse gear clutch oil pressure shall be not more than 5 p.s.i. below the main oil pressure.

3.3.2 Converter "out" oil temperature stabilizing. - When tested as specified in 4.6.3, the converter "out" oil temperature shall be stabilized for a minimum of 1 minute at 265° ± 5°F.

3.3.3 Service brakes. - When tested as specified in 4.6.4, the transmission input torque shall be within the applicable limits specified in table II.

Table II - Input torque (foot-pounds)

Type I		Type II	
Min.	Max.	Min.	Max.
666	759	545	629

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3.3.4 Right hand and left hand input speed and input torque. - When tested as specified in 4.6.5, the transmission input speed and input torque shall be within the applicable limits specified in table III.

Table III - Input speed and torque

Item	Type I Min.	Type I Max.	Type II Min.	Type II Max.
Input speed right and left hand r.p.m.	1,920	2,300	2,070	2,480
Input torque - foot-pounds	513	690	487	640

3.3.4.1 Lock-up clutch engagement. - When tested as specified in 4.6.6, the transmissions output speed shall be within the limits specified in table IV when the lock-up clutch engages.

3.3.4.2 Lock-up clutch disengagement. - When tested as specified in 4.6.6, the transmissions output speed shall be within the limits specified in table IV when the lock-up clutch disengages.

Table IV - Output speed

Item	Type I Min.	Type I Max.	Type II Min.	Type II Max.
Output speed with lock-up clutch engaged - r.p.m.	1,860	1,960	2,110	2,210
Output speed with lock-up clutch disengaged - r.p.m.	1,315	2,065	1,660	2,300

3.3.4.3 Right hand and left hand steer clutch pressure. - When tested as specified in 4.6.7, the right hand and left hand steer clutch pressure shall be not more than 5 p.s.i. below the main oil pressure.

3.3.4.4 Overspeed. - When tested as specified in 4.6.7, the overspeed limit pump pressure shall be a minimum of 9 p.s.i. above the lubricating oil pressure.

3.3.4.5 Right hand and left hand high gear steer clutch pressure. - When tested as specified in 4.6.8, the right hand and left hand steer clutch pressure shall be not more than 5 p.s.i. below the main oil pressure.

3.3.4.6 Right hand and left hand steer clutch output torque. - When tested as specified in 4.6.9, an output torque of 6,000 to 8,000 foot-pounds shall be required prior to clutch slip.

3.3.5 High gear input torque. - When tested as specified in 4.6.10, the high gear input torque shall be as specified in table V.

Table V. - High gear no load performance

Item	Type I Min.	Type I Max.	Type II Min.	Type II Max.
Input torque - foot-pounds		178		170
Main oil pressure p.s.i.	138	179	138	179
Converter "in" oil pressure p.s.i.		78		78
Main oil flow gallons per minute (GPM)	28		28	
Lubricating oil pressure p.s.i.	43	59	43	59
Converter "out" oil pressure p.s.i.		79		79
Governor oil pressure p.s.i.	140	180	140	180
Converter oil flow - GPM	12		12	

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3.3.5.1 High gear clutch oil pressure. - When tested as specified in 4.6.10, the minimum allowable clutch oil pressure in high gear shall be 5 p.s.i. below the main oil pressure.

3.3.5.2 High gear lock-up clutch. - When tested as specified in 4.6.10, the minimum allowable lock-up clutch oil pressure shall be 5 p.s.i. below the main oil pressure.

3.3.6 High gear mid-point. - When tested as specified in 4.6.11, the test results shall be within the limits specified in table VI.

Table VI. - High gear mid-point performance

Item	Types I and II	
	Min.	Max.
Converter "out" oil pressure p.s.i.	92	122
Converter "in" oil pressure p.s.i.	75	105
Main oil pressure - p.s.i.	138	179
Lubricating oil pressure p.s.i.	38	59
Main oil flow - GPM	24	29
Converter oil flow - GPM	34	43
Trans. input torque foot-pounds	*	**
Trans. output torque foot-pounds	*	**

See Acceptance Limits
*Fig. 1
**Fig. 2

3.3.7 High gear stall. - When tested as specified in 4.6.12, with the output shaft stalled, the test results shall be within the limits specified in table VII.

Table VII - High gear stall performance

Item	Types I and II	
	Min.	Max.
Main oil pressure - p.s.i.	138	179
Converter "in" oil pressure - p.s.i.	75	105
Lubricating oil pressure - p.s.i.	38	59
Main oil flow - GPM	24	
Converter oil flow - GPM	26	
Input speed - r.p.m.	*	**
Output torque - foot-pounds	*	**

See Acceptance Limits
*Fig. 1
**Fig. 2

3.3.8 Low gear, no load. - When tested as specified in 4.6.13, with no load, the test results shall be within the limits specified in table VIII.

Table VIII. - Low gear, no load performance

Item	Type I		Type II	
	Min.	Max.	Min.	Max.
Output speed - r.p.m.	722		660	
Main oil pressure - p.s.i.	138	179	138	179
Lubricating oil pressure - p.s.i.	45	66	45	66

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3.3.8.1 Low gear clutch oil pressure. - When tested as specified in 4.6.13, the clutch oil pressure (in low gear) shall be not more than 5 p.s.i. below the main oil pressure.

3.3.9 Low gear stall. - When tested as specified in 4.6.14, the test results of the main and lubricating oil pressure shall be within the applicable limits specified in table VIII.

3.3.10 Reverse gear mid-point. - When tested as specified in 4.6.15, at the reverse gear mid-point the test results of the main and lubricating oil pressure shall be within the applicable limits specified in table VII, and the input torque and output torque shall be within the limits specified in figure 1 or figure 2, as applicable.

3.3.10.1 Reverse gear clutch oil pressure. - When tested as specified in 4.6.15, the reverse gear clutch oil pressure shall be not more than 5 p.s.i. below the main oil pressure.

3.3.11 Reverse gear stall. - When tested as specified in 4.6.16, with the output shaft stalled, the main and lubricating oil pressure shall be within the applicable limits specified in table VII. The input speed and output torque shall be within the applicable limits specified in figure 1 and figure 2.

3.3.12 Neutral drag range. - When tested as specified in 4.6.17, the input torque shall be within 55 to 84 foot-pounds.

3.4 Leak check. - Pipe plugs shall be installed at pressure and temperature gage connections and all external surfaces wiped clean and dry. The transmission shall be operated as specified in 4.6.18, and checked for external leaks, in all gear ranges.

3.5 Temperature range. - When tested as specified in 4.6.19, the transmission shall operate satisfactorily throughout an ambient air temperature range from -65° to +125°F. The oil temperature in the transmission shall not exceed 300°F.

3.6 Identification markings. - The manufacturer's serial number shall be legibly and permanently stamped into the metal of each major transmission component housing, in a location visible when the transmission is installed in its intended vehicle. The marking shall be in accordance with Standard MIL-STD-130.

3.7 Workmanship. - Workmanship shall be such as to assure a product free of burs, rust, cracks, and chips, meeting the requiremens of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced therein.

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4.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Preproduction model. - A preproduction model selected from the first two units of production (see 3.1), shall be submitted for inspection and testing as specified herein. The model shall be representative of the transmission proposed to be furnished under the contract. Preproduction inspection and testing shall be conducted by the Government at a place designated by the Government, and shall consist of visual examination and all tests specified in 4.6 through 4.6.19.

4.3.1 Failure. - Failure of a preproduction model to pass any test specified herein shall be cause for refusal by the Government to conduct a retest until corrective measures, satisfactory to the Government, have been taken.

4.4 Acceptance inspection.

4.4.1 Sampling for transmissions.

4.4.1.1 Lot formation. - Unless otherwise specified, a lot shall consist of all transmissions of one part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.2 Sampling for acceptance examination. - Samples for acceptance examination shall be selected in accordance with Standard MIL-STD-105.

4.4.2 Examination inspection for transmissions.

4.4.2.1 Acceptable quality levels. - Each model selected in accordance with 4.4.1.2, shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.4.2.2 Classification of defects. - For examination purposes, defects shall be classified as follows:

Major defects (1.0 AQL)	Examination method STANDARD MEASURING EQUIPMENT
Dimensions out of tolerance affecting interchangeability of major components (see 3.2)	
Improper installation of hydraulic lines and fittings (see 3.2)	Visual
Improper linkage adjustment on controls (see 3.2)	Visual
Dirt, foreign matter in oil filter (see 3.2)	Visual
Oil leaks (see 3.4)	Visual
Malfunction of control levers and valves (see 3.3)	Visual & Functional

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Minor defects (2.5 AQL)	Examination method
Dimensions out of tolerance not affecting interchangeability of major components (see 3.2)	SME
Incorrect marking (see 3.6)	Visual
Faulty workmanship (see 3.7)	Visual

4.5 Acceptance tests. - Each transmission presented for acceptance shall be subjected to the tests specified in 4.6 through 4.6.18.

4.5.1 Failure. - Failure of a transmission to pass any test specified herein, shall be cause for rejection of that transmission. Rejected transmission may be resubmitted for acceptance after objective evidence has been provided by the manufacturer that corrective action has been taken to eliminate the conditions which caused rejection.

4.6 Test procedures.

4.6.1 Test conditions. - The transmission shall be installed on an absorption dynamometer test stand and all connections for input shafts, steering and shifting controls, oil lines, pressure and temperature outlets shall be made. The transmission output shafts shall be left uncoupled from the dynamometer. The transmission brakes shall be adjusted and the transmission speed increased to 1,000 ± 25 r.p.m., until main and converter oil pressures are stabilized. Tests shall be conducted at room temperature.

4.6.2 Pressure tests. - To determine conformance to 3.3.1, the transmission shall be operated as specified in 4.6.2.1 through 4.6.2.6, and the readings recorded.

4.6.2.1 Pressures. - The transmission shall be placed in neutral and the pressure recorded.

4.6.2.2 Right hand and left hand steer pressure. - To determine conformance to 3.3.1.2, the right hand steer clutch shall be applied, the pressure determined, and the clutch released. The left hand steer clutch shall then be applied, the pressure determined and the clutch released.

4.6.2.3 High gear clutch pressure. - To determine conformance to 3.3.1.3, the high gear clutch shall be engaged, the pressure determined and the clutch released.

4.6.2.4 Low gear clutch pressure. - To determine conformance to 3.3.1.4, the low gear clutch shall be engaged, the pressure determined and the clutch released.

4.6.2.5 Reverse gear clutch pressure. - To determine conformance to 3.3.1.5, the reverse gear clutch shall be engaged, the pressure determined and the clutch released.

4.6.2.6 Torque and pressure allowance tests. - To determine conformance to 3.3.1.1 the transmission shall be operated as specified in 4.6.2.1 and the following torques and pressures recorded:

(a) Input torque
(b) Main oil pressure
(c) Lubricating oil pressure
(d) Converter "in" oil pressure
(e) Converter "out" oil pressure

4.6.3 Converter "out" oil temperature stabilizing. - To determine conformance to 3.3.2, the transmission output shafts shall be coupled to the absorption dynamometer with no load on the dynamometer. The transmission shall be placed in low

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gear and the input speed increased to 2,300 ± 50 r.p.m., the service brakes shall then be applied with 240 ± 5 foot-pounds torque on the brake apply shaft. When the converter "out" oil temperature reaches 280°F., the transmission shall be placed in neutral position and the service brakes released, the transmission shall then be placed in low gear and allowed to run free until the converter "out" oil temperature is stabilized at 265° ± 5°F.

4.6.4 Service brakes. - To determine conformance to 3.3.3, the transmission output shafts shall be coupled to the absorption dynamometer with no load on the dynamometer. The transmission shall be placed in low gear and 240 foot-pounds torque applied to the service brakes to stall the output shafts, the input speed shall be increased to 2,300 r.p.m., and the input torque recorded.

4.6.5 Input speed and input torque. - To determine conformance to 3.3.4, the transmission output shafts shall be coupled to the absorption dynamometer, with no load on the dynamometer. The transmission shall be placed in neutral position and the service brakes applied with 240 foot-pounds torque on the right hand apply shaft and 190 foot-pounds torque on the left hand apply shaft. The transmission shall be placed in low gear and the input speed shall be increased until the left hand brake slips. When brake slippage occurs the input speed and the input torque shall be recorded. The above test shall be repeated, except that the right hand brake slippage shall be measured and the input torque and speed recorded.

4.6.6 Lock-up clutch. - To determine conformance to 3.3.4.1 and 3.3.4.2, the transmission output shafts shall be coupled to the absorption dynamometer, with no load on the dynamometer. The transmission service brakes shall be released and the transmission placed in high gear. When the converter "out" oil temperature reaches 265° ± 5°F., the input speed shall then be increased until the lock-up clutch engages. The output speed shall be recorded. The input speed shall be decreased until the lock-up clutch disengages, and the output speed again recorded.

4.6.7 Right hand and left hand steer clutch pressure. - To determine conformance to 3.3.4.3 and 3.3.4.4, the output shafts shall be coupled to the absorption dynamometer, with no load on the dynamometer. The transmission shall be placed in neutral position and the input speed increased to 2,000 r.p.m. When the converter "out" oil temperature stabilizes at 265° ± 5°F., the lubricating oil pressure shall be recorded. Full left steer shall be applied and the left hand steer clutch pressure recorded. The full left steer shall be released and the full right steer shall be applied. The right hand steer clutch pressure and the over-speed limit pump pressure shall then be recorded.

4.6.8 Right hand and left hand high gear steer clutch pressure. - To determine conformance to 3.3.4.5, the transmission output shafts shall be coupled to the absorption dynamometer, with the output shafts stalled. The transmission shall be placed in high gear and the input speed increased to 2,000 r.p.m. When the converter "out" oil temperature stabilizes at 265° ± 5°F., the main oil pressure shall be recorded. The full left steer shall be applied and the left hand steer clutch pressure shall then be recorded. The full left steer shall be released and the full right steer applied. The right hand steer clutch pressure shall then be recorded.

4.6.9 Right hand and left hand steer clutch output torque. - To determine conformance to 3.3.4.6, the transmission output shafts shall be coupled to the absorption dynamometer, with the output shafts stalled. The transmission shall be placed in neutral until the converter "out" oil temperature stabilizes at 265° ± 5°F. The right steer shall be applied with 90 p.s.i. oil pressure and the input speed increased until the clutch slips. The output torque shall then be recorded. The left steer shall be applied with 90 p.s.i. oil pressure and the input speed increased until the clutch slips. The output torque shall be recorded.

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4.6.10 High gear test. - To determine conformance to 3.3.5, 3.3.5.1 and 3.3.5.2, the transmission output shaft shall be coupled to the absorption dynamometer, with no load on the dynamometer. The transmission service brakes shall be released and the transmission placed in high gear, the input speed shall then be increased to 2,800 r.p.m. When the converter "out" oil temperature reaches 260° ± 20°F., the pressure, oil flow, and torque shall be recorded.

4.6.11 High gear mid-point. - To determine conformance to 3.3.6, the transmission output shafts shall be coupled to the absorption dynamometer. The transmission service brakes shall be released and the transmission placed in high gear, the input speed shall then be increased to 2,350 r.p.m. The absorption dynamometer shall be loaded to produce 1,150 r.p.m., output speed for type I, or to produce 1,070 r.p.m. output speed for type II. When the converter "out" oil temperature reaches 260° ± 20°F., the pressure and oil flow shall be recorded.

4.6.12 High gear stall. - To determine conformance to 3.3.7, the transmission output shafts shall be coupled to the absorption dynamometer. The tranmission service brakes shall be released and the transmission placed in high gear. The output shafts shall be stalled and 750 foot-pounds input torque shall then be applied for type I or 610 foot-pounds input torque for type II. When the converter "out" oil temperature reaches 260° ± 20°F., the pressure and oil flow shall be recorded.

4.6.13 Low gear no load. - To determine conformance to 3.3.8 and 3.3.8.1, the transmission output shafts shall be coupled to the absorption dynamometer, with no load on the dynamometer. The transmission service brakes shall be released and the transmission placed in low-gear, the input speed shall then be increased to 2,800 r.p.m. When the converter "out" oil temperature reaches 260° ± 20°F., the pressure and oil flow shall be recorded.

4.6.14 Low gear stall. - To determine conformance to 3.3.9, the transmission output shafts shall be coupled to the absorption dynamometer, with no load on the dynamometer. The transmission shall be placed in low gear and the transmission service brakes applied to stall the output shafts. An input torque of 750 foot-pounds shall be applied for type I, or 610 foot-pounds for type II. When the converter "out" oil temperature reaches 260° ± 20°F., the pressure shall be recorded.

4.6.15 Reverse gear mid-point. - To determine conformance to 3.3.10 and 3.3.10.1, the transmission output shafts shall be coupled to the absorption dynamometer. The transmission service brake shall be released and the transmission placed in reverse gear. The input speed shall then be increased to 2,250 r.p.m. The absorption dynamometer shall be loaded to produce, 300 r.p.m. output speed for type I, or 280 r.p.m. output speed for type II. When the converter "out" oil temperature reaches 260° ± 20°F., the pressure and oil flow shall be recorded.

4.6.16 Reverse gear stall. - To determine conformance to 3.3.11, the transmission output shafts shall be coupled to the absorption dynamometer. The transmission shall be placed in reverse gear and the transmission service brakes shall then be applied to stall the output shafts. An input torque of 750 foot-pounds shall be applied for type I, or 610 foot-pounds shall be applied for type II. When the converter "out" oil temperature reaches 260° ± 20°F., the pressure and oil flow shall be recorded.

4.6.17 Neutral drag range. - To determine conformance to 3.3.12, the transmission output shafts shall be coupled to the absorption dynamometer, with full excitation applied on the absorption dynamometer. The transmission service brakes shall be released and the transmission placed in neutral range, the input speed shall then be increased to 2,000 r.p.m. When the converter "out" oil temperature reaches 260° ± 20°F., the input torque shall be recorded.

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4.6.18 Leak check. - To determine conformance to 3.4, the transmission output shafts shall be coupled to the absorption dynamometer with no load on the dynamometer. The transmission service brakes shall be released and the input speed increased to 2,100 r.p.m. When the converter "out" oil temperature reaches 290° ± 10°F., the transmission shall be checked for external leaks, in all gear ranges.

4.6.19 Extreme temperature test. - To determine conformance to 3.5, the transmission shall be serviced with the applicable oil (see 3.2.1) and placed in an environmental chamber at a temperature of minus 65° ± 5°F., for a period of 24 hours. The transmission shall be coupled to the absorption dynamometer and all output shafts, steering and shifting controls, oil lines, pressure and temperature outlets connected. With the ambient temperature remaining at minus 65° ± 5°F., the transmission input speed shall be slowly increased to 1000 r.p.m. until main and converter oil pressures are stabilized. All steering and shifting positions shall be observed for any indication of noise, drag or malfunction. The transmission shall then be serviced with the applicable oil (see 3.2.1) and the above procedure shall be repeated except that the temperature shall be raised to plus 125° ± 5°F. The converter "out" oil temperature shall not exceed 300°F.

4.7 Inspection of preparation for delivery. - Inspection of preservation, packaging, packing and marking, shall be in accordance with the applicable procedures specified in Specification MIL-T-10693 (see 5.1).

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - Preservation, packaging, packing, marking and the inspection and testing thereof shall be in accordance with the applicable requirements of Specification MIL-T-10693, for the level of protection specified (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of preproduction model required (see 3.1).
(c) Selection of the applicable level of protection of the referenced specification (see 5.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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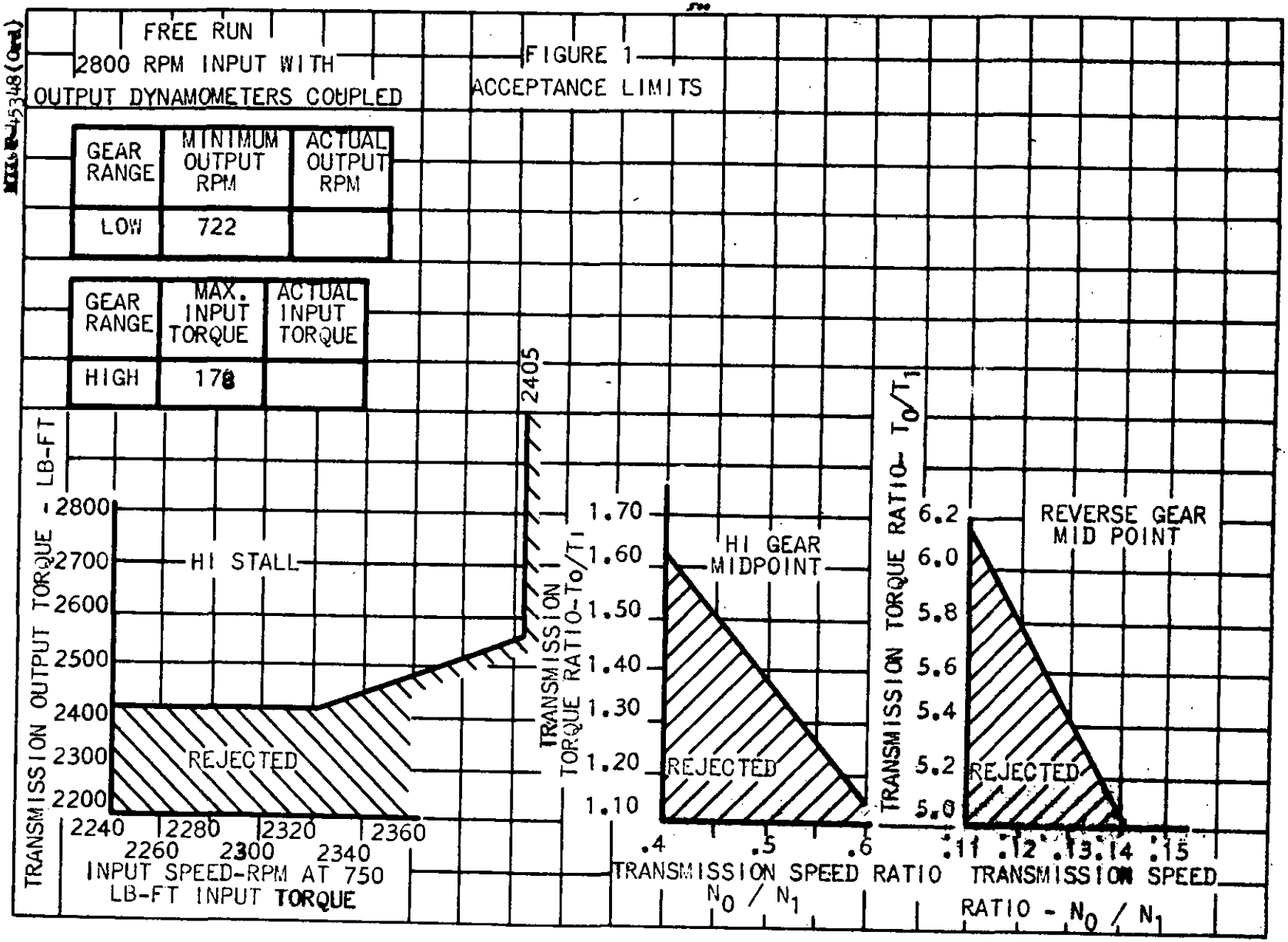
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FREE RUN
2800 RPM INPUT WITH
OUTPUT DYNAMOMETERS COUPLED

FIGURE 1
ACCEPTANCE LIMITS

GEAR RANGE	MINIMUM OUTPUT RPM	ACTUAL OUTPUT RPM
LOW	722	

GEAR RANGE	MAX. INPUT TORQUE	ACTUAL INPUT TORQUE
HIGH	178	



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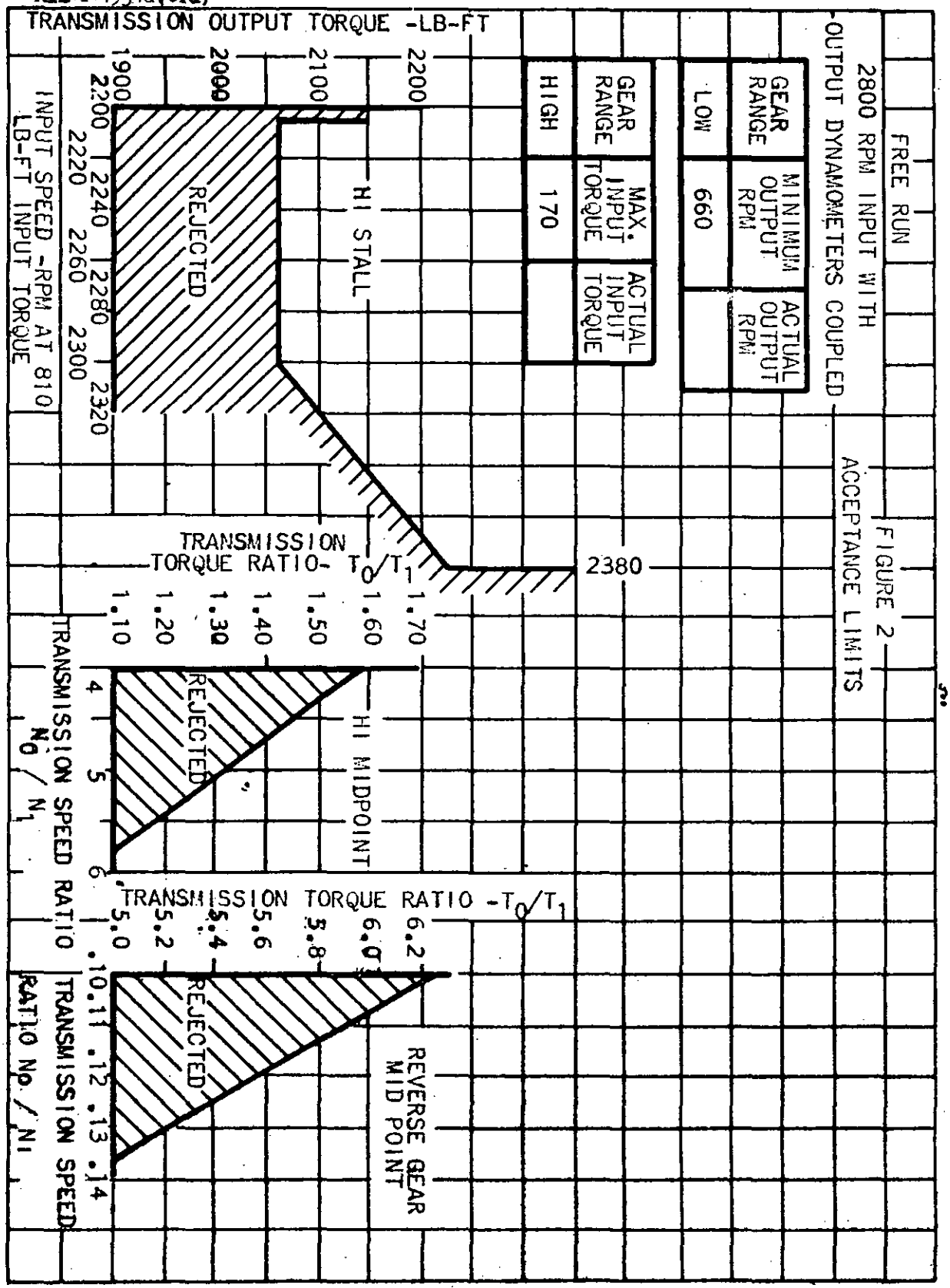


FIGURE 2

GPO 886903

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